

V3 empirical study · Instruction

Seven models · four underlyings · four volatility regimes · 1.5-year monthly calibration · GARCH-Merton Q-map up

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of 7 American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Ranking column ranks 1 = best.

Experimental design (held fixed for every model)

- Models: GBM, MD-GBM (Markov Directional Geometric Brownian Motion), GARCH(1,1), Heston (no jumps), Merton jump-diffusion, GARCH-Merton, Heston-Merton (Bates).
- MD-GBM means Markov Directional Geometric Brownian Motion. In this study it is specified in V4_MD-GBM_model.pdf in this folder.
- Companies: SPY (primary), AAPL, MSFT, AMZN. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- Calibration window: 1.5-year ending at each monthly update. Rolling: monthly (period start plus month-ends). Parameters estimated at t are used only after t (no look-ahead).
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

V3 empirical study · Instruction (continued)

Seven models · four underlyings · four volatility regimes · 1.5-year monthly calibration · GARCH-Merton Q-map up

Shared market sample and filters (not model-specific)

- For each company $\times$ regime, one listed-call sample is drawn once and reused by all seven models in this report: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston-Merton calibration and to the LSM comparison set: (1) calls only, finite S , K , C ; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \text{DTE} \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05 , valid bid-ask with relative spread $\leq 50\%$ when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH-Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston-Merton: $(\kappa, \theta, \xi, \rho, v_0)$ are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \rightarrow r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t ; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff-Schwartz American call on the full path cloud ($n_{\text{paths}} = 10000$, seed 42, $n_{\text{steps}} = \text{DTE}$, $\Delta t = 1/252$). Paths are not averaged before stopping.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE\%} = 100 \times \text{mean}(|C_{\text{model}} - C_{\text{mkt}}|/C_{\text{mkt}})$. $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Positive bias = model expensive vs market.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all seven models in this report)

Regime	Window	SPY n	AAPL n	MSFT n	AMZN n
2008-2009 Crisis	2008-08-01 → 2009-07-31	50	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-31	50	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-30	50	51	50	51
2019-2020 COVID	2019-09-01 → 2020-08-31	50	50	50	50

Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7-60	S/K-1 ≤10%	Liquidity
SPY	2008-2009	3359	3311	3311	3203	3203
SPY	2013-2014	12815	12548	12548	12405	12405
SPY	2018-2019	35222	35104	35104	34809	34809
SPY	2019-2020	42100	41961	41961	41379	41379
AAPL	2008-2009	685	685	685	652	652
AAPL	2013-2014	8925	8875	8875	8633	8633
AAPL	2018-2019	6309	6297	6297	6084	6084
AAPL	2019-2020	8010	7992	7992	7713	7713
MSFT	2008-2009	566	566	566	538	538
MSFT	2013-2014	3299	3237	3237	3175	3175
MSFT	2018-2019	7814	7752	7752	7583	7583
MSFT	2019-2020	6892	6852	6852	6655	6655
AMZN	2008-2009	465	465	465	442	442
AMZN	2013-2014	5715	5692	5692	5598	5598
AMZN	2018-2019	28809	28808	28808	28346	28346
AMZN	2019-2020	26766	26760	26760	26323	26323

*Input is the processed call panel restricted to (regime end – 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with $\text{trading_date} \leq t$. LSM prices the Monday contract using the latest calibration row with date $\leq$ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid-ask mid when the source mark is \$0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All seven models in this report still share that exact sample.

Heston / Heston-Merton calibration further subsamples the filtered lookback surface to ≤ 24 contracts (maturity $\times$ moneyness strata). GBM / Merton / GARCH / GARCH-Merton estimate from lookback log returns only (jumps: 3σ residual threshold where applicable).

Page 4 · Table block · SPY · seven models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	55.01	40.97	+21.16	7
MD-GBM	41.02	32.65	-4.71	3
GARCH	43.04	35.72	+33.41	4
Heston	25.90	22.28	+13.29	1
Merton	51.73	37.42	+25.72	6
GARCH-Merton	44.31	37.49	+36.30	5
Heston-Merton	38.05	33.61	+25.36	2

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	38.08	30.45	+24.88	5
MD-GBM	29.99	24.17	+13.83	1
GARCH	36.35	29.04	+22.86	2
Heston	59.26	47.06	+44.22	6
Merton	37.32	29.83	+23.72	3
GARCH-Merton	37.90	30.58	+24.87	4
Heston-Merton	74.75	66.23	+65.36	7

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	49.51	41.57	+32.90	2
MD-GBM	36.48	31.07	+14.03	1
GARCH	65.98	49.13	+41.26	5
Heston	49.97	41.27	+36.18	3
Merton	60.60	52.40	+46.39	4
GARCH-Merton	68.12	51.50	+44.06	6
Heston-Merton	70.07	59.97	+58.12	7

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	64.88	57.43	+44.13	5
MD-GBM	43.63	36.23	+14.85	1
GARCH	53.62	42.08	+35.24	4
Heston	47.54	40.64	+29.57	2
Merton	70.92	64.09	+53.08	6
GARCH-Merton	52.76	41.90	+35.13	3
Heston-Merton	72.56	65.60	+56.44	7

Page 5 · Table block · AAPL · seven models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	57.67	42.83	+31.10	7
MD-GBM	43.89	32.72	+12.84	5
GARCH	34.36	28.77	+25.46	3
Heston	31.07	25.81	+19.95	1
Merton	55.37	40.83	+36.40	6
GARCH-Merton	40.33	34.50	+32.14	4
Heston-Merton	33.89	28.49	+23.08	2

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	50.20	39.70	+38.63	6
MD-GBM	30.36	23.26	+17.13	1
GARCH	40.08	29.28	+23.86	3
Heston	34.83	27.66	+27.17	2
Merton	48.87	38.01	+37.02	5
GARCH-Merton	59.86	50.37	+50.37	7
Heston-Merton	40.21	33.41	+33.36	4

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	32.53	28.14	+18.76	2
MD-GBM	23.64	20.27	+5.37	1
GARCH	40.82	29.83	+21.75	5
Heston	34.04	28.08	+25.17	3
Merton	36.65	31.82	+28.29	4
GARCH-Merton	54.71	44.78	+41.77	7
Heston-Merton	42.20	35.75	+34.85	6

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	46.19	39.52	+30.76	5
MD-GBM	31.97	25.38	+9.95	1
GARCH	45.53	33.84	+29.35	4
Heston	34.22	28.97	+23.49	2
Merton	47.03	40.14	+35.87	6
GARCH-Merton	57.26	46.26	+44.06	7
Heston-Merton	41.76	36.21	+31.48	3

Page 6 · Table block · MSFT · seven models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	44.02	34.76	+17.99	5
MD-GBM	34.43	27.73	+3.32	3
GARCH	50.93	41.05	+35.87	6
Heston	25.98	23.13	+9.45	1
Merton	41.80	34.10	+23.08	4
GARCH-Merton	61.38	51.11	+48.59	7
Heston-Merton	29.00	26.08	+15.28	2

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	62.49	53.14	+52.98	5
MD-GBM	38.79	31.25	+28.02	1
GARCH	85.94	73.78	+73.69	6
Heston	40.25	33.10	+32.34	2
Merton	56.13	47.20	+46.76	4
GARCH-Merton	108.47	97.36	+97.36	7
Heston-Merton	50.19	42.58	+42.11	3

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	41.73	34.38	+27.55	5
MD-GBM	25.61	21.67	+4.03	1
GARCH	37.25	30.23	+24.44	3
Heston	27.61	21.05	+15.81	2
Merton	43.31	35.57	+33.24	6
GARCH-Merton	48.79	40.81	+38.95	7
Heston-Merton	40.75	34.60	+32.94	4

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	45.93	37.46	+28.12	4
MD-GBM	31.75	23.91	+4.32	1
GARCH	49.20	34.48	+25.36	5
Heston	33.07	26.92	+19.27	2
Merton	50.06	41.56	+34.80	6
GARCH-Merton	54.17	40.27	+34.17	7
Heston-Merton	43.27	37.93	+32.57	3

Page 7 · Table block · AMZN · seven models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	46.92	36.28	+22.05	7
MD-GBM	35.59	28.09	-0.26	5
GARCH	27.67	23.47	+10.57	2
Heston	26.60	21.97	+17.33	1
Merton	46.51	35.73	+31.26	6
GARCH-Merton	34.76	29.27	+26.15	4
Heston-Merton	30.92	26.21	+21.89	3

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	31.09	25.49	+16.54	2
MD-GBM	21.30	19.23	-0.63	1
GARCH	34.57	27.93	+19.87	4
Heston	32.30	24.22	+19.81	3
Merton	35.62	26.86	+18.58	5
GARCH-Merton	57.06	45.50	+44.68	7
Heston-Merton	36.36	28.31	+24.81	6

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	43.66	38.02	+25.21	5
MD-GBM	34.59	30.76	+12.20	3
GARCH	32.44	27.42	+19.15	2
Heston	29.98	23.82	+17.85	1
Merton	49.34	43.76	+35.22	7
GARCH-Merton	44.74	37.34	+33.22	6
Heston-Merton	36.66	30.79	+28.23	4

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
GBM	52.42	42.24	+31.41	6
MD-GBM	40.03	33.48	+14.41	3
GARCH	30.12	25.22	+15.17	1
Heston	36.37	30.91	+19.68	2
Merton	59.46	51.55	+42.61	7
GARCH-Merton	46.47	38.78	+31.83	5
Heston-Merton	45.25	39.09	+30.63	4

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells best
1	MD-GBM	33.94	34.51	10
2	Heston	35.56	33.55	5
3	GARCH	44.24	40.45	1
4	Heston-Merton	45.37	41.25	0
5	GBM	47.65	46.56	0
6	Merton	49.42	49.10	0
7	GARCH-Merton	54.44	53.47	0

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
GBM	0	0	0	0	0
MD-GBM	3	3	3	1	10
GARCH	0	0	0	1	1
Heston	1	1	1	2	5
Merton	0	0	0	0	0
GARCH-Merton	0	0	0	0	0
Heston-Merton	0	0	0	0	0

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
GBM	50.90	45.47	41.86	52.35	47.65
MD-GBM	38.73	30.11	30.08	36.85	33.94
GARCH	39.00	49.23	44.12	44.61	44.24
Heston	27.39	41.66	35.40	37.80	35.56
Merton	48.85	44.48	47.47	56.87	49.42
GARCH-Merton	45.20	65.82	54.09	52.67	54.44
Heston-Merton	32.96	50.38	47.42	50.71	45.37

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
GBM	51.87	52.26	0
MD-GBM	37.78	38.75	3
GARCH	49.75	48.33	0
Heston	45.67	48.75	1
Merton	55.14	56.16	0
GARCH-Merton	50.77	48.54	0
Heston-Merton	63.86	71.32	0

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
GBM	46.65	48.20	0
MD-GBM	32.47	31.17	3
GARCH	40.20	40.45	0
Heston	33.54	34.13	1
Merton	46.98	47.95	0
GARCH-Merton	53.04	55.99	0
Heston-Merton	39.51	40.98	0

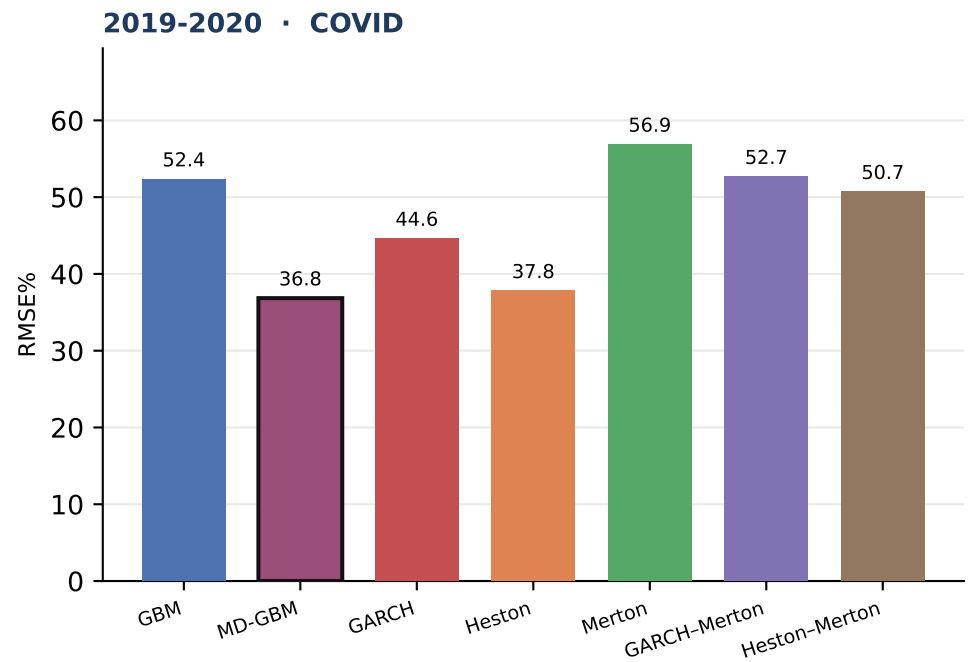
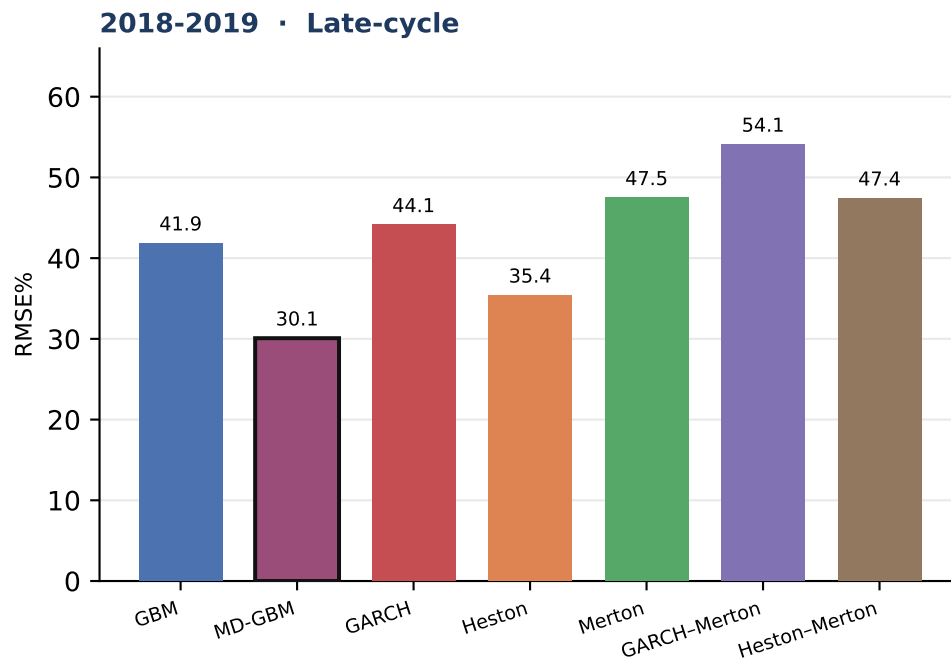
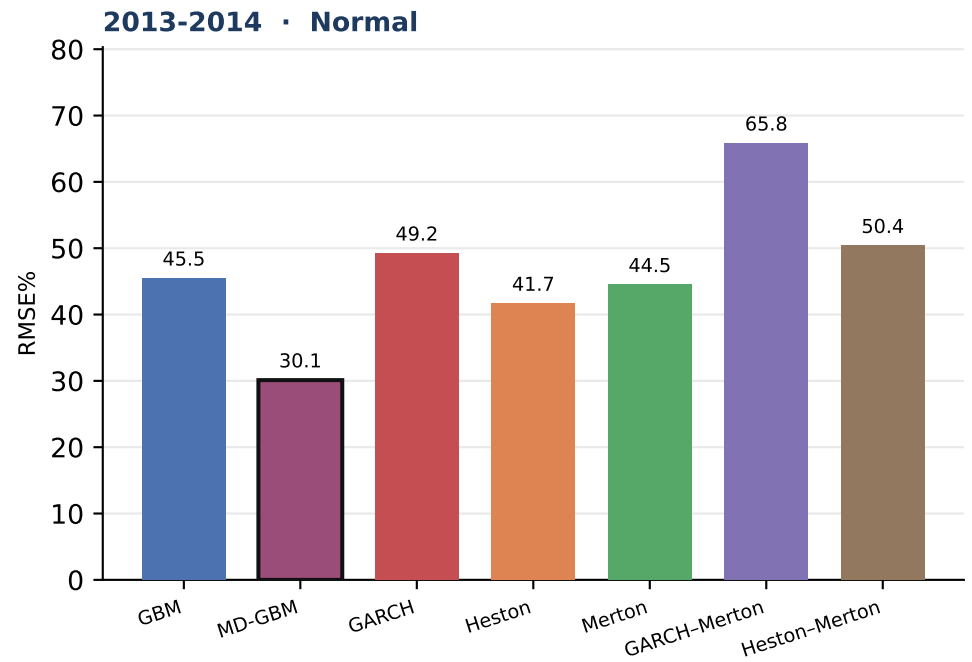
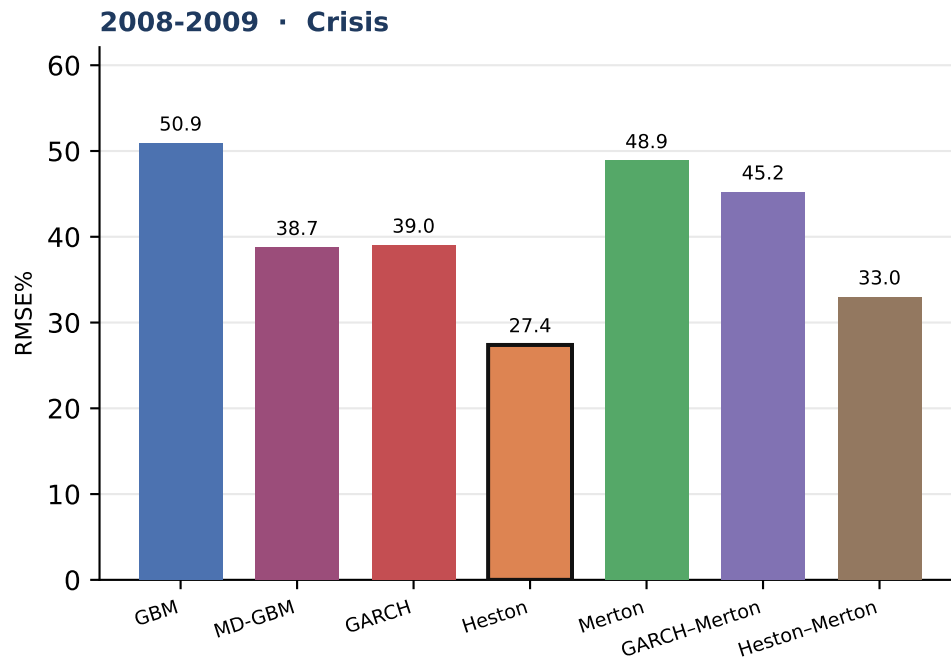
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
GBM	48.54	44.98	0
MD-GBM	32.65	33.09	3
GARCH	55.83	50.06	0
Heston	31.73	30.34	1
Merton	47.83	46.69	0
GARCH-Merton	68.20	57.78	0
Heston-Merton	40.80	42.01	0

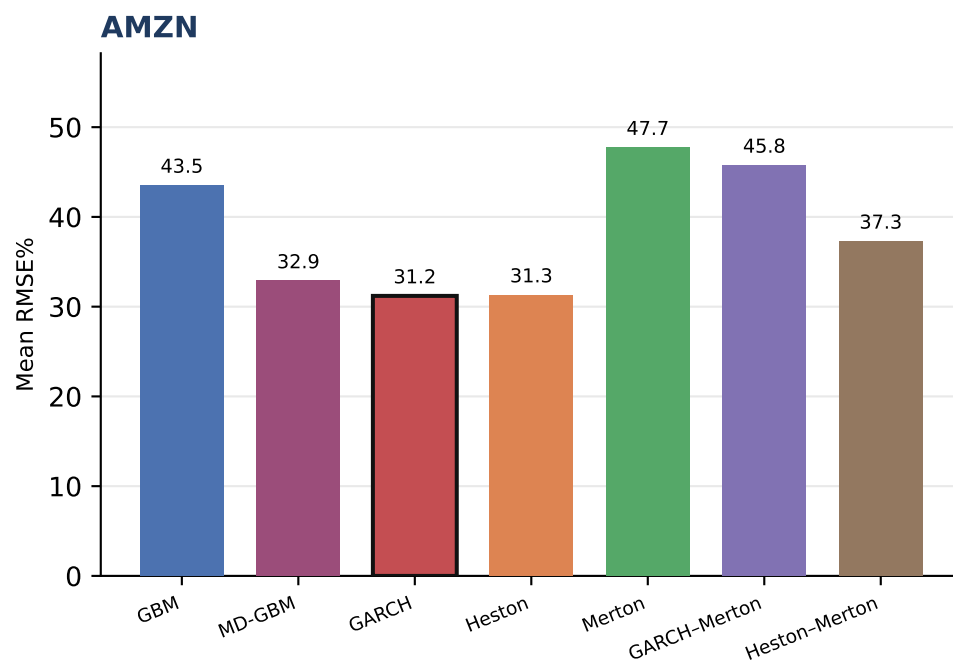
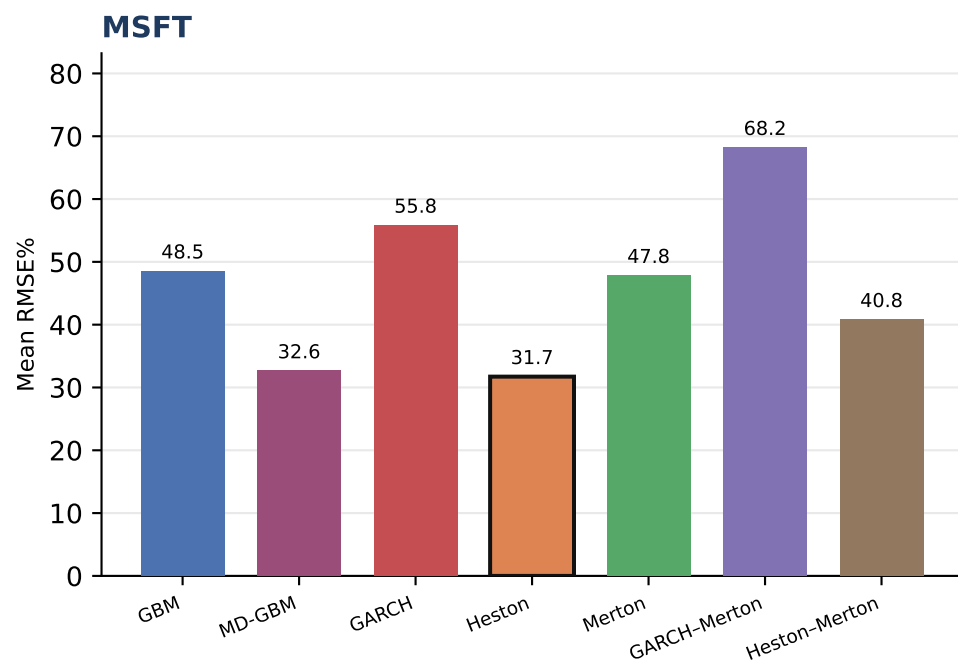
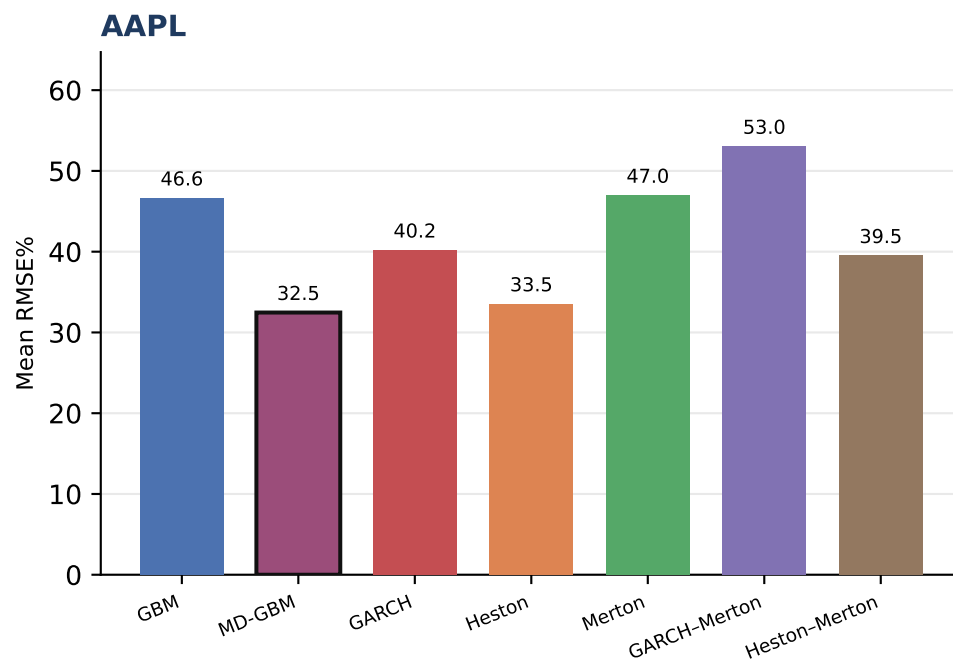
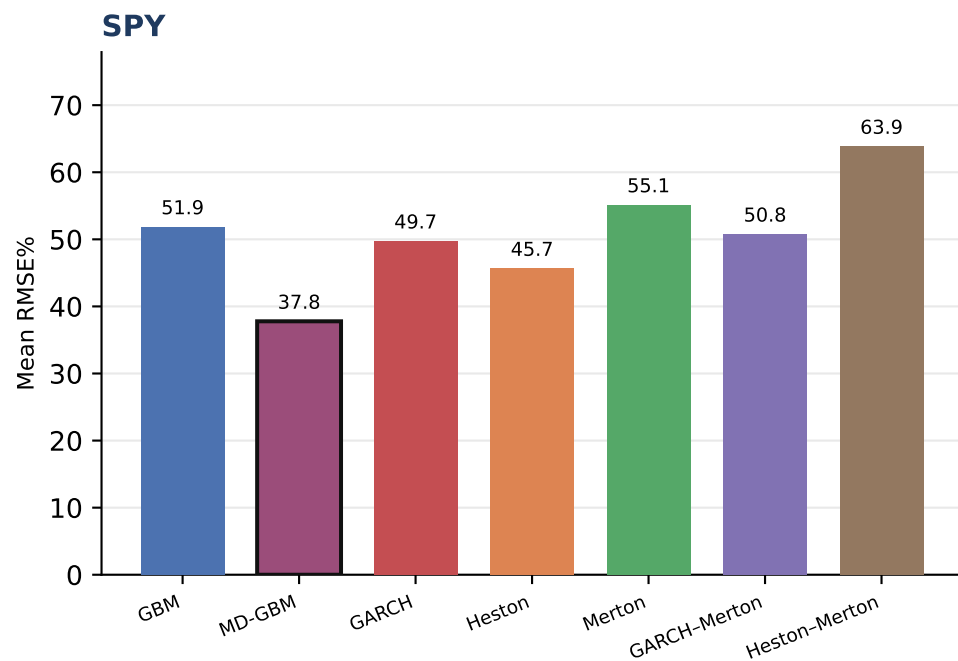
Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
GBM	43.52	45.29	0
MD-GBM	32.88	35.09	1
GARCH	31.20	31.28	1
Heston	31.31	31.14	2
Merton	47.73	47.93	0
GARCH-Merton	45.76	45.60	0
Heston-Merton	37.30	36.51	0

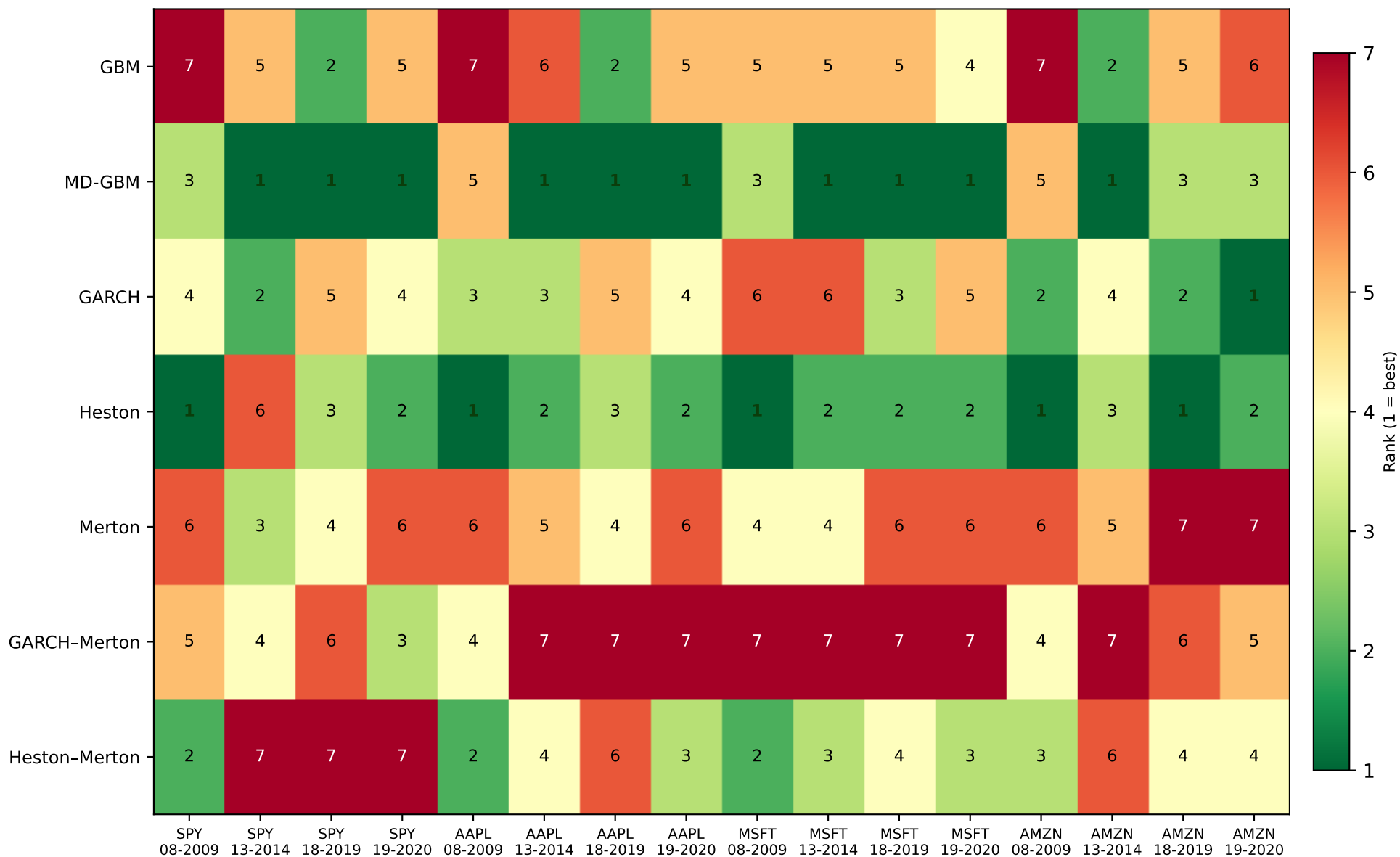
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



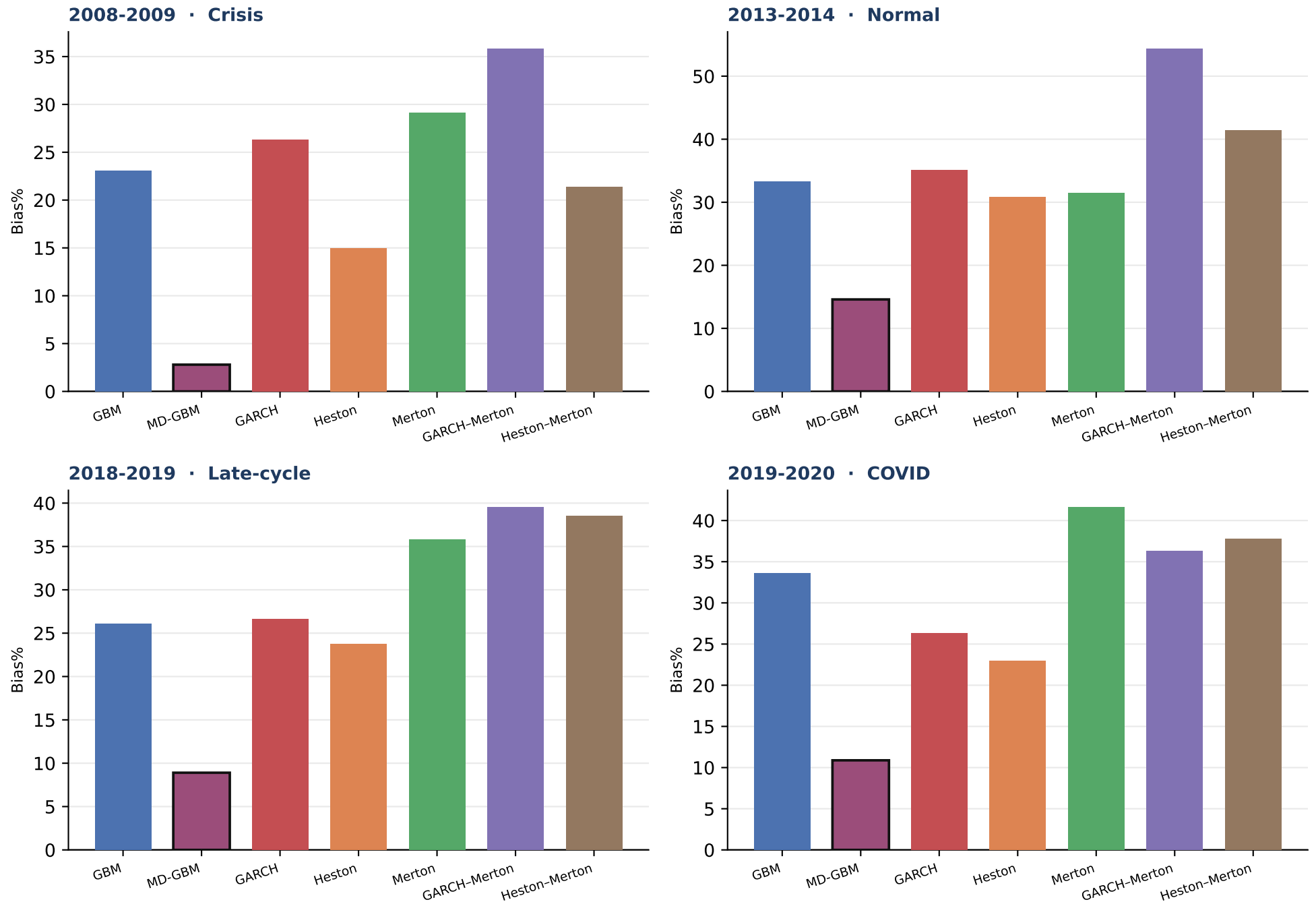
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero Bias% (not the RMSE ranking). Positive = model expensive vs listed quote.

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 16 company×regime cells, MD-GBM is the best overall model (mean RMSE% = 33.94; 10 of 16 cells).
- Second overall is Heston (mean RMSE% = 35.56; 5 cells).
- The ranking is not stable. Best-in-cell models across the 16 tables: GARCH, Heston, MD-GBM.
- SPY: lowest mean RMSE% is MD-GBM (37.78); wins 3 of 4 regimes.
- AAPL: lowest mean RMSE% is MD-GBM (32.47); wins 3 of 4 regimes.
- MSFT: lowest mean RMSE% is Heston (31.73); wins 1 of 4 regimes.
- AMZN: lowest mean RMSE% is GARCH (31.20); wins 1 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is Heston (27.39).
- 2013-2014 (Normal): best mean RMSE% is MD-GBM (30.11).
- 2018-2019 (Late-cycle): best mean RMSE% is MD-GBM (30.08).
- 2019-2020 (COVID): best mean RMSE% is MD-GBM (36.85).
- No-jump models as a group have lower mean RMSE% (40.35) than the jump models in this report (49.74).
- For the overall winner (MD-GBM), crisis RMSE% is 38.73 vs 30.11 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year monthly calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1...MSFT-4 uses exactly the same option contracts for all seven models in this report. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE% and Bias% are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).